

2 (a)	$\rho = m/V$ $V = (\pi d^2/4) \times t = 7.67 \times 10^{-7} \text{ m}^3$ $\rho = (9.6 \times 10^{-3})/[\pi(22.1/2 \times 10^{-3})^2 \times 2.00 \times 10^{-3}]$ $\rho = 12513 \text{ kg m}^{-3}$ (allow 2 or more s.f.)	C1	
		C1	
		A1	[3]
(b) (i)	$\Delta\rho/\rho = \Delta m/m + \Delta t/t + 2\Delta d/d$ $= 5.21\% + 0.50\% + 0.905\%$ [or correct fractional uncertainties] $= 6.6\%$ (6.61%)	C1	
		C1	
		A1	[3]
(ii)	$\rho = 12500 \pm 800 \text{ kg m}^{-3}$	A1	[1]